Xiang Li

Postdoctoral Researcher · Yazhouwan National Laboratory · Sanya, Hainan, China

lixiang117423@gmail.com · GitHub · Google Scholar · Blog

I am a postdoctoral researcher in the Crop Disease Resistance Mechanisms and Technology Team at Yazhouwan National Laboratory, working with Jianmin Zhou. My work uses genome and population-scale sequencing to locate disease-resistance genes in crops and their wild relatives: I built PlantNLRatlas, a dataset of NLR resistance genes across 100 chromosome-level plant genomes, and I am currently building a wild soybean (*Glycine soja*) germplasm and pan-genome resource to mine resistance loci and genes. I received my PhD in plant pathology from Yunnan Agricultural University in 2025, including a research stay at INRAE's Plant Health Institute of Montpellier, and I maintain open-source analysis tools such as qPCRtools, ggmotif, bioRtools and biopytools.

Positions

- 2025-06—Present **Postdoctoral Researcher**, Yazhouwan National Laboratory — Crop Disease Resistance Mechanisms and Technology Team; advisor: Jianmin Zhou
- 2022-03—2022-12 **Visiting PhD Researcher**, INRAE — Plant Health Institute of Montpellier — Host: Jean-Benoit Morel

Education

- 2021-09—2025-06 **PhD in Plant Pathology**, Yunnan Agricultural University — Dissertation: Study on Mechanism of Intra-specific Interactions Inducing Plant Resistance Among Acuce Pure Lines (advisors: Youyong Zhu and Huichuan Huang)
- 2018-09—2021-06 **MSc in Plant Pathology**, Yunnan Agricultural University
- 2014-09—2018-06 **BSc in Plant Protection**, Yunnan Agricultural University

Awards

- 2024 **Youyong Scholarship**, Yunnan Agricultural University
- 2022—2023 **First-Class University Scholarship**, Yunnan Agricultural University — Awarded in both 2022 and 2023

Publications

1. De Mita, Stéphane, Duan, Shengchang, **Li, Xiang**, Pélissier, Rémi, Mathieu, Laura, Li, Jingyu, Ma, Linna, Li, Zihang, Liao, Jingjing, He, Xiahong, Zhu, Youyong, Li, Chengyun, Dong, Yang, Morel, Jean-Benoit, Huang, Huichuan. **Population genomics of the rice landrace Acuce reveals exceptional dynamics of immune receptors**. *Plant Diversity*, 2026. (co-first author) · doi:10.1016/j.pld.2026.08.010
2. Li, Feixue, Ma, Linna, Li, Jingyu, Ye, Chen, Mei, Xinyue, Zhao, Siji, Li, Yiwen, Liu, Yixiang, Huang, Huichuan, **Li, Xiang**. **Context-dependent reshaping of rhizosphere microbial communities by intercropping systems: A systematic analysis of 32 crop combinations**. *Rhizosphere*, 38, 101368, 2026. (corresponding author) · doi:10.1016/j.rhisph.2026.101368

3. Ma, Linna, Liu, Yemei, Li, Feixue, Bai, Yusha, Li, Yingguang, Chen, Xiaoqi, Huang, Huichuan, **Li, Xiang**. **Network analysis reveals that bHLH transcription factors positively regulate rice resistance to Magnaporthe oryzae**. *BMC Plant Biology*, 26, 1717, 2026. (corresponding author) · doi:10.1186/s12870-026-09237-8

4. Ma, Linna, **Li, Xiang**, Zhang, Zhiping, Zhang, Ting, Duan, Haibin, Huang, Huichuan, Liu, Yixiang, Zhu, Shusheng, Zhu, Youyong, Li, Yingbin. **Soil microbial community and chemical properties response to blueberry—soybean intercropping system**. *Plant and Soil*, 507(1), 881—895, 2025. (co-first author) · doi:10.1007/s11104-024-06775-8

5. Ma, Linna, Bai, Yusha, **Li, Xiang**, Li, Jingyu, Li, Feixue, Yue, Yan, Li, Yingguang, Guo, Jingwen, Ye, Chen, Mei, Xinyue, Du, Fei, Liu, Yixiang, Yang, Min, Zhu, Shusheng, Huang, Huichuan. **Compartment-specific core microbiomes and functions in rice**. *Plant and Soil*, 514(2), 2939—2958, 2025. (co-first author) · doi:10.1007/s11104-025-07558-5

6. De Mita, Stéphane, Duan, Shengchang, **Li, Xiang**, Pélissier, Rémi, Mathieu, Laura, Li, Jingyu, Ma, Linna, Li, Zihang, Liao, Jingjing, He, Xiahong, Zhu, Youyong, Li, Chengyun, Dong, Yang, Morel, Jean-Benoit, Huang, Huichuan. **Population genomics of the rice landrace Acuce reveals exceptional dynamics of immune receptors**. *bioRxiv*, 2024. (co-first author) · doi:10.1101/2024.02.15.579491

7. **Li, Xiang**, Ma, Linna, Wang, Yingmin, Ye, Chen, Guo, Cunwu, Li, Yingbin, Mei, Xinyue, Du, Fei, Huang, Huichuan. **PlantNLRatlas: a comprehensive dataset of full- and partial-length NLR resistance genes across 100 chromosome-level plant genomes**. *Frontiers in Plant Science*, 14, 1178069, 2023. (first author) · doi:10.3389/fpls.2023.1178069

8. **Li, Xiang**, Ma, Linna, Mei, Xinyue, Liu, Yixiang, Huang, Huichuan. **ggmotif: An R Package for the extraction and visualization of motifs from MEME software**. *PLOS ONE*, 17(11), e0276979, 2022. (first author) · doi:10.1371/journal.pone.0276979

9. **Li, Xiang**, Wang, Yingmin, Li, Jingyu, Mei, Xinyue, Liu, Yixiang, Huang, Huichuan. **qPCRtools: An R package for qPCR data processing and visualization**. *Frontiers in Genetics*, 13, 1002704, 2022. (first author) · doi:10.3389/fgene.2022.1002704

Software

- **bioRtools** (R) — An R toolkit for routine bioinformatics and plotting. github.com/lixiang117423/bioRtools
- **biopytools** (Python) — A Python command-line toolkit for routine bioinformatics tasks. github.com/lixiang117423/biopytools
- **DeepBSA-rs** (Rust) — Deep-learning BSA (bulked segregant analysis) QTL mapping reimplemented in pure Rust as a single binary with no runtime dependencies. github.com/lixiang117423/deepbsa-rs
- **qPCRtools** (R) — An R package for qPCR data processing and visualization. github.com/lixiang117423/qPCRtools
- **ggmotif** (R) — An R package for extracting and visualizing motif information from MEME output. github.com/lixiang117423/ggmotif